2(a)	mass × velocity	**B1**
2(b)(i)	kinetic energy $= \frac{1}{2}mv^2$	**C1**
	$= \frac{1}{2} \times 0.24 \times 2.3^2$	**C1**
	$= 0.63\,\text{J}$	**A1**
2(b)(ii)	change in momentum $= \frac{1}{2} \times 240 \times 5.0 \times 10^{-3}$	**C1**
	$= 0.60\,\text{N s}$	**A1**
2(b)(iii)	(change in velocity of Y) $= 0.60 / 0.12$ $(= 5.0\,\text{m s}^{-1})$	**C1**
	final velocity of Y $= 5.0 - 2.3$ $= 2.7\,\text{m s}^{-1}$	**A1**
	or	
	(final momentum of Y) $= 0.60 - 0.12 \times 2.3$ $(= 0.324\,\text{N s})$	**(C1)**
	final velocity of Y $= 0.324 / 0.12$ $= 2.7\,\text{m s}^{-1}$	**(A1)**
2(c)	sloping straight line from (0, 0) to $t = 3.0\,\text{ms}$ **and** another straight line continuous with the first from $t = 3.0\,\text{ms}$ to (5.0, 0)	**B1**
	lines showing maximum force of magnitude 240 N	**B1**
	lines wholly in the negative F region of the graph	**B1**